# AIDA BEHMARD

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## RESEARCH INTERESTS

Exoplanets, galactic archaeology, stellar astrophysics, statistics, machine learning and data-centric techniques, large surveys

## APPOINTMENTS

**Flatiron Research Fellow** 2024–present
Center for Computational Astrophysics, *New York, NY*

**Kalbfleisch Postdoctoral Fellow** 2023–2024
American Museum of Natural History, *New York, NY*

**Post-Baccalaureate Fellow** 2015–2017
Princeton University, *Princeton, NJ*

## EDUCATION

**California Institute of Technology**, Pasadena, CA 2017–2023
Advisor: Prof. Heather Knutson

**Ph.D.** Planetary Science
**M.S.** Planetary Science

**Yale University**, New Haven, CT 2011–2015
**B.S.** Physics

## HONORS & AWARDS

Research Corporation for Science Advancement Fellow 2026
Block Award, Aspen Center for Physics 2023
*A prized award given to one promising junior physicist during each winter conference*
Caltech 3-Minute Thesis Competition – $1^{st}$ Place 2022
NASA ExoExplorers 2022
Keck Institute for Space Studies Affiliate 2019
NSF Graduate Research Fellowship 2018–2021
Origins of Life Initiative Grant, Harvard University 2015
Science, Technology, and Research Scholars (STARS II) Program 2014
George J. Schulz Fellowship for the Physical Sciences 2013
Yale College Dean's Undergraduate Research Fellowship 2012

## AWARDED TELESCOPE TIME

ESPRESSSO/VLT – 6 hours awarded, FTC 2026 (**PI**)
STIS/HST Cycle 34 – 28 orbits awarded, 2026-2027 (Co-I)
Magellan/Clay: MIKE – 4 nights awarded, 2025-2026 (Co-I)
Gemini North 8.1m Observatory: MAROON-X – 6.67 hours awarded, FT 2025 (**PI**)
WIYN 3.5m Observatory: NEID – 10 hours awarded, 2024B (**PI**)

Keck Observatory: HIRES – 1 night awarded, 2021A (**Sci PI**)
Keck Observatory: HIRES – 1 night awarded, 2020B (**Sci PI**)
Hubble Space Telescope – 2 nights awarded (Co-I), 2016

## CONFERENCE TALKS

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| Cool Stars 23, *Tokyo, Japan* | June 2026 |
| NY Area Exoplanet Meeting, *New York, NY* | May 2026 |
| Rocky Worlds 4, *Groningen, Netherlands* | Jan. 2026 |
| Know Thy Star, Know Thy Planet 2, *Pasadena, CA* | Feb. 2025 |
| Two HoRSEs, *Berlin, Germany* | July 2024 |
| Cool Stars 22, *San Diego, CA* | June 2024 |
| Gordon Research Conference, *South Hadley, MA* | June 2023 |
| Late-Stage Exoplanet Systems, Aspen Center for Physics, *Aspen, CO* | Mar. 2023 |
| Exoplanet Demographics, *virtual* | Nov. 2020 |
| Extreme Precision in Radial Velocity IV, *Grindelwald, Switzerland* | Mar. 2019 |
| Keck Science Meeting, *Pasadena, CA* | Sept. 2018 |
| Exoplanets in Southern California IV, *Pasadena, CA* | Sept. 2018 |

## SEMINARS & COLLOQUIA

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| Astronomy Seminar, U of Utah | May 2026 |
| Astrophysics Seminar, U of Pennsylvania | Oct. 2025 |
| Exoplanets & Stars Seminar, Yale University | April 2025 |
| Astronomy Seminar, Universidad Diego Portales | April 2025 |
| Astronomy Seminar, Vanderbilt University | Nov. 2024 |
| SDSS-V Galactic Genesis Working Group Meeting | Oct. 2023 |
| Exoplanet Seminar, NASA Goddard | Jan. 2023 |
| TESS Science Talk Series, MIT | Nov. 2022 |
| APS Seminar, CU Boulder | Oct. 2022 |
| Planetary Science Seminar, UCLA | Sept. 2022 |
| Astronomy Colloquium, AMNH | Sept. 2022 |
| Astronomy Lunch Talk, Columbia University | Sept. 2022 |
| ESPF Seminar Series, STScI | Aug. 2022 |
| NASA ExoExplorers Science Series | June 2022 |
| Exoplanet Journal Club, NASA JPL | April 2022 |
| CEHW Seminar, Penn State | Feb. 2022 |
| EPL Astronomy Seminar, Carnegie Observatories | Feb. 2022 |
| TESS Science Team Meeting #27 | Jan. 2022 |
| Exoplanet Meeting, Princeton University | Nov. 2021 |
| FLASH Seminar, UC Santa Cruz | Dec. 2020 |
| Tea Talk, Carnegie Observatories | Dec. 2018 |
| Origins of Life Research Symposium, Harvard University | Aug. 2015 |

## TEACHING & MENTORING

### Teaching Assistant

Held office hours, wrote problem set solutions, graded homework and exams, and substituted for instructor on multiple occasions

Ay/Ge 117: *Bayesian Statistics and Data Analysis* Winter 2020, 2021, 2022
Ay/Ge 133: *Formation & Evolution of Planetary Systems* Spring 2019

**Guest Lectures**
Dartmouth College, ASTR 19: *Habitable Planets* Spring 2026

**Research Mentoring**

- Haniyeh Tajer — CCA Pre-Doc Program, Dec. 2025–present
  *Data-Driven Chemical Abundances for Gaia RVS Spectra* — Ohio State U Ph.D. student
- Chris Lam — April 2025–present
  *Data-Driven Asteroseismic Ages for Solar-like Stars* — U of Florida Ph.D. → Caltech/IPAC Postdoc
- Zoe Hacksaw — CCA Pre-Doc Program, Jan.–June 2026
  *Mapping Neutron-Capture Elements Across Galactic Components* — UT Austin Ph.D. student
- Daija Ricks — Simons-NSBP Fellow, May–Aug. 2025
  *Exploring the Milky Way with Data-Driven Stellar Abundances* — NC Central U undergrad
- Cinta Vidante — AMNH Kade Fellow, May–July 2024
  *Rotationally-Modulated Flares in Young Stars with ML* — U of Potsdam M.S. → MPIA Ph.D. student
- Cassie Sevilla — Caltech SURF Fellow, June 2021–Aug. 2022
  *Li Abundance Signatures Following Planet Engulfment* — Caltech undergrad → Cornell Ph.D. student
  Publication: C. Sevilla et al. (2022), *MNRAS*, 516, 3

## PROFESSIONAL SERVICE

† dates redacted for anonymity

NSF NOIRLab TAC†
NASA Review Panel (x2)†
CCA Workshop "Planets in the Galactic Context" Lead Organizer 2026
Committee On INclusiveness in SDSS (COINS) Member 2024–present
Referee for *AAS Journals, MNRAS, Nature* 2019–present
ExoNYC Conference Organizer 2025
Dix Caltech Planetary Science Seminar Organizer 2020–2021
Caltech Stars and Planets Astro-ph Organizer 2019–2020

## SELECTED OUTREACH

**NASA ExoExplorers Alumni Mentor** 2025–present
NASA ExoPAG & Exoplanet Exploration Program

**Pasadena Unified Volunteer K-2nd Science Teacher** 2017–2023
Caltech Center for Teaching, Learning, and Outreach

**Buddy Program Mentor** 2021–2022
Caltech Geological and Planetary Sciences Division

**Summer Internship Mentor** 2019–2021
Caltech WAVE Program

**Diversity Committee Chair** 2018–2021
Caltech Graduate Student Council

**Classroom Instructor** 2020
Skype a Scientist

**Graduate Student Mentor** 2017–2020
Caltech Women Mentoring Women Program

### Invited Outreach Talks

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| Caltech Seminar Day | 2022 |
| Invited speaker, Los Altos High School Physics Club | 2021 |
| Caltech FUTURE of Physics conference | 2021 |
| Women of Aeronautics and Astronautics India Chapter | 2021 |
| NorCal/Nevada American Association of Physics Teachers Meeting | 2021 |

## PUBLICATIONS

*h-index* = 25, *total citations* = 2123, [full list on ADS]
1st/2nd-author ($\underline{\star}$ directly supervised student, † co-first authors):

1. **A. Behmard** et. al (2025), "A Link Between Rocky Planet Densities and Host Star Elemental Abundances", *The Astronomical Journal*, 170, 282
2. **A. Behmard** et. al (2025), "A Data-Driven M Dwarf Model and Detailed Abundances for ∼17,000 M Dwarfs in SDSS-V", *The Astrophysical Journal*, 982, 13
3. S. Vissapragada, **A. Behmard**† (2025), "The Hottest Neptunes Orbit Metal-Rich Stars", *The Astronomical Journal*, 169, 2
4. **A. Behmard**, E. Cunningham, M. Bedell, M. Ness (2023), "Elemental Abundances of *Kepler* Objects of Interest in APOGEE DR17", *The Astronomical Journal*, 165, 178
5. **A. Behmard**, F. Dai, J. Brewer, T. Berger, A. Howard (2023), "Planet Engulfment Detections are Rare According to Observations and Stellar Modeling", *MNRAS*, 521, 2
6. **A. Behmard**, C. Sevilla, J. Fuller (2023), "Planet Engulfment Signatures in Twin Stars", *MNRAS*, 518, 4
7. C. Sevilla$\underline{\star}$, **A. Behmard**, J. Fuller (2022), "Long-Term Lithium Abundance Signatures Following Planetary Engulfment", *MNRAS*, 516, 3
8. **A. Behmard**, F. Dai, A. Howard (2022), "Stellar Companions To TESS Objects of Interest: A Test of Planet-Companion Alignment", *The Astronomical Journal*, 163, 160
9. **A. Behmard**, E. Petigura, A. Howard (2019), "Data-Driven Spectroscopy of Cool Stars at High Spectral Resolution", *The Astrophysical Journal*, 876, 68

10. **A. Behmard**, D. Graninger, E. Fayolle, J. Bergner, K. Öberg (2019), "Desorption Kinetics and Binding Energies of Small Hydrocarbons", *The Astrophysical Journal*, 875, 73

Nth-author:

1. SDSS Collaboration et al. [including **A. Behmard**] (2026),"The Nineteenth Data Release of the Sloan Digital Sky Survey", *The Astrophysical Journal Supplement Series*, 285, 1
2. A. Sinha et al. [including **A. Behmard**] (2026),"The chemical homogeneity of single-lined spectroscopic binaries in open clusters", *MNRAS*, 547, 4
3. J. Kollmeier et al. [including **A. Behmard**] (2025),"Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy", *The Astronomical Journal*, 171, 1
4. F. Wanderley et al. [including **A. Behmard**] (2025),"An Analysis of the Radius Gap in a Sample of Kepler, K2, and TESS Exoplanets Orbiting M-dwarf Stars", *The Astrophysical Journal*, 993, 233
5. A. Howard et al. [including **A. Behmard**] (2025),"Planet Masses, Radii, and Orbits from NASA's K2 Mission", *The Astrophysical Journal Supplement Series*, 278, 2
6. J. Van Zandt et al. [including **A. Behmard**] (2025),"The TESS–Keck Survey. XXIV. Outer Giants May Be More Prevalent in the Presence of Inner Small Planets ", *The Astronomical Journal*, 169, 235
7. M. Greklek-McKeon et al. [including **A. Behmard**] (2025),"Tidally Heated Sub-Neptunes, Refined Planetary Compositions, and Confirmation of a Third Planet in the TOI-1266 System", *The Astronomical Journal*, 169, 6
8. J. Galarza et al. [including **A. Behmard**] (2025),"HIP 8522: A Puzzling Young Solar Twin with the Lowest Detected Lithium Abundance", *The Astrophysical Journal*, 983, 1
9. Y. Lu et al. [including **A. Behmard**] (2025),"Evidence of Truly Young High-$\alpha$ Dwarf Stars", *The Astronomical Journal*, 169, 3
10. R. Rubenzahl et al. [including **A. Behmard**] (2024),"KPF Confirms a Polar Orbit for KELT-18 b", *The Astronomical Journal*, 168, 5
11. H. Isaacson et al. [including **A. Behmard**] (2024),"The California Legacy Survey. V. Chromospheric Activity Cycles in Main-sequence Stars", *The Astrophysical Journal Supplement Series*, 274, 2
12. D. Pidhorodetska et al. [including **A. Behmard**] (2024),"The TESS-Keck Survey. XXII. A Sub-Neptune Orbiting TOI-1437", *The Astronomical Journal*, 168, 3
13. A. Polanski et al. [including **A. Behmard**] (2024),"The TESS-Keck Survey. XX. 15 New TESS Planets and a Uniform RV Analysis of All Survey Targets", *The Astrophysical Journal Supplement Series*, 272, 2

14. S. Lange et al. [including **A. Behmard**] (2024),"The TESS-Keck Survey. VII. A Superdense Sub-Neptune Orbiting TOI-1824", *The Astronomical Journal*, 167, 6

15. B. Hord et al. [including **A. Behmard**] (2024),"Identification of the top TESS objects of interest for atmospheric characterization of transiting exoplanets with JWST", *The Astronomical Journal*, 167, 5

16. R. Rubenzahl et al. [including **A. Behmard**] (2024),"The TESS-Keck Survey. XII. A Dense 1.8 R Ultra-Short-Period Planet Possibly Clinging to a High-Mean-Molecular-Weight Atmosphere After the First Gyr", *The Astronomical Journal*, 167, 4

17. M. Hill et al. [including **A. Behmard**] (2024),"The TESS–Keck Survey. XIX. A Warm Transiting Sub-Saturn-mass Planet and a Nontransiting Saturn-mass Planet Orbiting a Solar Analog", *The Astronomical Journal*, 167, 4

18. C. Beard et al. [including **A. Behmard**] (2024),"The TESS-Keck Survey. XVII. Precise Mass Measurements in a Young, High-multiplicity Transiting Planet System Using Radial Velocities and Transit Timing Variations", *The Astronomical Journal*, 167, 2

19. J. Murphy et al. [including **A. Behmard**] (2023),"The TESS-Keck Survey. XVI. Mass Measurements for 12 Planets in Eight Systems", *The Astronomical Journal*, 166, 4

20. S. Blunt et al. [including **A. Behmard**] (2023),"Overfitting Affects the Reliability of Radial Velocity Mass Estimates of the V1298 Tau Planets", *The Astronomical Journal*, 166, 2

21. M. MacDougall et al. [including **A. Behmard**] (2023),"The TESS-Keck Survey. XV. Precise Properties of 108 TESS Planets and Their Host Stars", *The Astronomical Journal*, 166, 1

22. C. Brinkman et al. [including **A. Behmard**] (2023),"TOI-561 b: A Low-density Ultra-short-period "Rocky" Planet around a Metal-poor Star", *The Astronomical Journal*, 165, 88

23. J. Van Zandt et al. [including **A. Behmard**] (2023),"TESS-Keck Survey. XIV. Two Giant Exoplanets from the Distant Giants Survey", *The Astronomical Journal*, 165, 60

24. F. Dai et al. [including **A. Behmard**] (2023),"TOI-1136 is a Young, Coplanar, Aligned Planetary System in a Pristine Resonant Chain", *The Astronomical Journal*, 165, 33

25. M. El Mufti et al. [including **A. Behmard**] (2022),"TOI 560: Two Transiting Planets Orbiting a K Dwarf Validated with iSHELL, PFS, and HIRES RVs", *The Astronomical Journal*, 165, 1

26. M. MacDougall et al. [including **A. Behmard**] (2022),"The TESS-Keck Survey. XIII. An Eccentric Hot Neptune with a Similar-Mass Outer Companion around TOI-1272", *The Astronomical Journal*, 164, 97

27. A. Chontos et al. [including **A. Behmard**] (2022),"The TESS-Keck Survey: Science Goals and Target Selection", *The Astronomical Journal*, 163, 297

28. E. Petigura et al. [including **A. Behmard**] (2022),"The California-Kepler Survey. X. The Radius Gap as a Function of Stellar Mass, Metallicity, and Age", *The Astronomical Journal*, 163, 179

29. J. Winters et al. [including **A. Behmard**] (2022),"A Second Planet Transiting LTT 1445A and a Determination of the Masses of Both Worlds", *The Astronomical Journal*, 163, 61

30. P. Dalba et al. [including **A. Behmard**] (2022),"The TESS-Keck Survey. VIII. Confirmation of a Transiting Giant Planet on an Eccentric 261 Day Orbit with the Automated Planet Finder Telescope", *The Astronomical Journal*, 163, 61

31. N. Heidari et al. [including **A. Behmard**] (2022),"HD 207897 b: A dense sub-Neptune transiting a nearby and bright K-type star", *Astronomy & Astrophysics*, 658, A176

32. J. Murphy et al. [including **A. Behmard**] (2021),"Another Superdense Sub-Neptune in K2-182 b and Refined Mass Measurements for K2-199 b and c", *The Astronomical Journal*, 162, 294

33. M. MacDougall et al. [including **A. Behmard**] (2021),"The TESS-Keck Survey. VI. Two Eccentric Sub-Neptunes Orbiting HIP-97166", *The Astronomical Journal*, 162, 265

34. A. Polanski et al. [including **A. Behmard**] (2021),"Wolf 503 b: Characterization of a Sub-Neptune Orbiting a Metal-Poor K Dwarf", *The Astronomical Journal*, 162, 238

35. N. Scarsdale et al. [including **A. Behmard**] (2021),"TESS-Keck Survey. V. Twin Sub-Neptunes Transiting the Nearby G Star HD 63935", *The Astronomical Journal*, 162, 215

36. M. Rice et al. [including **A. Behmard**] (2021),"SOLES I: The Spin-Orbit Alignment of K2-140 b, *The Astronomical Journal*, 162, 182

37. F. Dai et al. [including **A. Behmard**] (2021),"TKS X: Confirmation of TOI-1444b and a Comparative Analysis of the Ultra-short-period Planets with Hot Neptunes", *The Astronomical Journal*, 162, 62

38. B. Fulton et al. [including **A. Behmard**] (2021), " California Legacy Survey. II. Occurrence of Giant Planets beyond the Ice Line ", *The Astrophysical Journal Supplement Series*, 255, 14

39. L. Rosenthal et al. [including **A. Behmard**] (2021), " The California Legacy Survey. I. A Catalog of 178 Planets from Precision Radial Velocity Monitoring of 719 Nearby Stars over Three Decades ", *The Astrophysical Journal Supplement Series*, 255, 8

40. L. Weiss et al. [including **A. Behmard**] (2021), "The TESS-Keck Survey II: Masses of Three Sub-Neptunes Transiting the Galactic Thick-Disk Star TOI-561", *The Astronomical Journal*, 161, 2

41. M. Kosiarek et al. [including **A. Behmard**] (2020),"Physical Parameters of the Multi-Planet Systems HD 106315 and GJ 9827", *The Astronomical Journal*, 161, 1

42. F. Dai et al. [including **A. Behmard**] (2020), "The TESS-Keck Survey III: An aligned orbit for TOI-1726 c", *The Astronomical Journal*, 160, 4

43. R. Cloutier et al. [including **A. Behmard**] (2020), "TOI-1235 b: a keystone super-Earth for testing radius valley emergence models around early M dwarfs", *The Astronomical Journal*, 160, 22

44. P. Dalba et al. [including **A. Behmard**] (2020), "The TESS-Keck Survey I: A Warm Sub-Saturn-mass Planet and a Caution about Stray Light in TESS Cameras", *The Astronomical Journal*, 159, 5

45. E. Gaidos et al. [including **A. Behmard**] (2019), "Planetesimals Around Stars with *TESS* (PAST): I. Transient Dimming of a Binary Solar Analog at the End of the Planet Accretion Era", *MNRAS*, 488, 4465

46. M.C.Y. Lau, R. Harris, Y. Oh, M. Joo Yi, **A. Behmard**, T.C. Onstott (2018), "Taxonomic and functional compositions impacted by the quality of metatranscriptomic assemblies", *FEMS Microbiology Ecology*, 9, 1235